Networks and the defenders who watch them



Cyber security · Level 4 · Lesson 3

In this lesson you will

- Explain packets, addresses and routes.
- Name the three security goals and spot which one a problem breaks.
- Describe layers of defence for a real network.

Key ideas

- Packets hop through routers by many routes
- Confidentiality, integrity, availability
- Layers: firewall, separate networks, updates, backups
- People watch the logs and alarms

Activity: Defend the network

1. Warm-up (3 min): A photo’s journey
2. Lab (6 min): Keep the messages flowing
3. Map (6 min): Layers of defence
4. Share (3 min): One improvement

Cut four ‘packets’ from paper, numbered 1 to 4. No real passwords or network names on the maps.

Think it through

1. Why does the internet keep working when parts of it break?
2. Which goal does a leaked class list break? A changed mark? A crashed website?
3. Why is one strong defence not enough?

Before you go

- One layer of defence, and the goal it protects.